

## Chapter 2: Arithmetic Expressions

1. Identify the odd one out from the following expressions, after INTERCHANGING their operators.

a)  $204 + 3 \div 29$

b)  $67 - 2 \times 37$

c)  $48 \div 6 \times 3$

d)  $195 \times 49 - 2$

**Answer:** c

**Solution:**

We have to identify the odd one out after interchanging the operators present in the expressions.

Let us solve each equation one by one:

**Option a:**  $204 + 3 \div 29$

After interchanging the operators, the resultant expression is:  $204 \div 3 + 29$

$$= 68 + 29$$

$$= 97$$

**Option b:**  $67 - 2 \times 37$

After interchanging the operators, the resultant expression is:  $67 \times 2 - 37$

$$= 134 - 37$$

$$= 97$$

**Option c:**  $48 \div 6 \times 3$

After interchanging the operators, the resultant expression is:  $48 \times 6 \div 3$

$$= 288 \div 3$$

$$= 96$$

**Option d:**  $195 \times 49 - 2$

After interchanging the operators, the resultant expression is:  $195 - 49 \times 2$

$$= 195 - 98$$

$$= 97$$

In every option, the answer is 97 except in option c, where the resultant value is 96.

Hence, the correct answer is option c.

2. There is a seminar from 11:00 AM to 2:00 PM, where students must spend AT LEAST 1 hour in the seminar. The table displays the number of students entering and exiting the seminar at different times. What is the MAXIMUM possible number of students who spent at least 2 hours in the seminar?

| SR. NO | TIME     | IN  | OUT |
|--------|----------|-----|-----|
| 1.     | 11:00 AM | 120 |     |
| 2.     | 12:00 PM | 40  | 60  |
| 3.     | 1:00 PM  | 80  | 40  |
| 4.     | 2:00 PM  |     | 140 |

a) 60

b) 80

c) 100

d) 120

**Answer:** c

**Solution:**

To maximize the number of students spending at least 2 hours in the seminar, we should aim to retain as many students as possible for that duration.

1. Initial Entry at 11:00 AM:

- 120 students entered the seminar
- 60 students exited at 12:00 PM, meaning these 60 students cannot complete 2 hours
- At 12:00 PM, the remaining 60 students from the 11:00 AM batch are still in the hall

2. At 12:00 PM:

- 40 new students entered the hall
- The total number of students in the seminar is now 100 (60 from the 11:00 AM batch + 40 new students)
- The 60 students from the 11:00 AM batch have now completed 1 hour

3. At 1:00 PM:

- 80 more students entered
- 40 students left the hall, and to maximize the number staying for 2 hours, we assume these 40 students are from the 11:00 AM batch, meaning they have now completed 2 hours
- The remaining 20 students from the 11:00 AM batch stay in the hall
- The total number of students in the hall is now 140 (20 from 11:00 AM + 40 from 12:00 PM + 80 from 1:00 PM)

4. At 2:00 PM:

- No new students enter, and all 140 students exit
- By this time, the 40 students from the 12:00 PM batch have completed 2 hours

Final Calculation:

- 60 students from the 11:00 AM batch stayed until at least 1:00 PM, and 40 of them left exactly at 1:00 PM after completing 2 hours
- 20 students from the 11:00 AM batch remained until 2:00 PM, so they also completed 2 hours
- 40 students from the 12:00 PM batch stayed until 2:00 PM, completing 2 hours

Thus, the maximum possible number of students who spent at least 2 hours in the seminar is 100. Option c is correct.

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3. In the given expression ' $8 @ 7 - 15 < 4 \times 2 \times 5$ ', which operator cannot be used in place of '@', such that the condition remains valid?

- a) -                                      b)  $\div$                                       c) +                                      d)  $\times$

**Answer:** d

**Solution:**

$$8 @ 7 - 15 < 4 \times 2 \times 5$$

The left side of the condition has  $8 @ 7 - 15$  and the right side has  $4 \times 2 \times 5$ , which is 40.

Let's solve it logically.

The LHS has -15 in it. So, whatever the value of  $8 @ 7$  be, its value further decreases due to the subtraction of 15 from it.

So, Subtraction (-) and division ( $\div$ ) reduce the value, and there is already a (-15) in the expression, which decreases the value further.

So, the result will definitely be less than 40, and these operators can be used definitely.

Addition (+) increases the value slightly. Let's check with "+".

$8 + 7 - 15 = 0$ , which is still less than 40. So, addition can be used here.

Multiplication ( $\times$ ) increases the value.

$8 \times 7 - 15 = 41$ , which is not less than 40. So, multiplication cannot be used.

Therefore, the operator that cannot be used is multiplication ( $\times$ ). Option d is correct.

4. A bus has 40 seats, and some seats are already occupied, with one passenger in each seat.

- At the first stop, 5 passengers get off and 8 passengers get in
- At the second stop, 2 passengers get off and 4 passengers get in
- At the third stop, exactly half of the passengers get off

After this, half of the bus seats are occupied.

Which of the following expressions represents the number of passengers on the bus at the beginning?

a)  $5 \times 5 + 5$

b)  $6 \times 6 - 4$

c)  $5 \times 5 + 10$

d)  $6 \times 6 - 5$

**Answer:** c

**Solution:**

This can be solved by moving backwards from the last stop. (reverse deduction)

Half of the bus seats are occupied at the end.

Since the bus has 40 seats, 20 passengers remain. ( $40/2 = 20$ )

At the third stop, half of the passengers got down.

This means before the third stop, there were 40 passengers, because half of 40 is 20.

At the second stop, 2 passengers got down and 4 got in.

If there are 'X' number of passengers and 2 got down and 4 got in, the expression for the number of people at that time is:

$$X - 2 + 4$$

$$= X + 2$$

This means the number of passengers increased by 2 (overall).

So, before the second stop there were 38 passengers only, which later became 40.

At the first stop, 5 passengers got down and 8 got in.

If there are 'Y' number of passengers and 5 passengers got down and 8 got in, the expression for the number of people at that time is:

$$Y - 5 + 8$$

$$= Y + 3$$

This means the number of passengers increased by 3 (overall), so at the beginning, there were 35 passengers. ( $35 + 3 = 38$ )

Among the given expressions,

a)  $5 \times 5 + 5 = 25 + 5 = 30$

b)  $6 \times 6 - 4 = 36 - 4 = 32$

c)  $5 \times 5 + 10 = 25 + 10 = 35$

d)  $6 \times 6 - 5 = 36 - 5 = 31$

Therefore, the correct answer is option c.

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5. The following expression has a whole number 'n'. If a blank can either take '×4' or '−12', what is the least possible value of 'n'?

**Expression:**  $(n \text{ \_\_\_\_ } \text{ \_\_\_\_ } ) \text{ \_\_\_\_ } = 96$

a) 9

b) 18

c) 4

d) 72

**Answer:** a

**Solution:**

Think logically.

Multiplying by 4 makes the number grow bigger and subtracting 12 makes it smaller.

If we want the smallest starting number 'n' and turn it into 96, we should multiply by 4 more times, because even a small number can become big after multiplying. As subtraction decreases a number, let's try to avoid it as much as possible.

Now, the expression must make n into 96.

Let's assume that all three operations performed on n are of '×4' only.

$$\text{So, } n \times 4 \times 4 \times 4 = 96$$

$$n \times 64 = 96.$$

But, there is no such whole number which when multiplied by 64 gives 96 as the result.

Thus, we can say that not all the three operations are '×4'. At least one '−12' must be used.

Now let's try using two '×4' operations and one '−12' operation, since we want to subtract as little as possible. There are different possible arrangements. Let's test them carefully:

**Case 1: Subtract first, then multiply twice**

$$(n - 12 \times 4) \times 4 = 96$$

Divide by 4 on both sides.

$$(n - 12 \times 4) = 24$$

$$n - 48 = 24$$

Add 48 on both sides.

$$n = 24 + 48$$

$$n = 72$$

This is a whole number.

**Case 2: Multiply once, subtract, then multiply**

$$(n \times 4 - 12) \times 4 = 96$$

$$(4n - 12) \times 4 = 96$$

$$16n - 48 = 96$$

Add 48 on both sides.

$$16n = 144$$

$$n = 9$$

This is also a whole number.

**Case 3: Multiply twice, then subtract**

$$(n \times 4 \times 4) - 12 = 96$$

$$16n - 12 = 96$$

Add 12 on both sides.

$$16n = 108$$

$$n = 108/16 = 6.75$$

$$n = 6.75$$

It is not a whole number.

Now, compare the valid whole numbers:

$$\text{Case 1: } n = 72$$

$$\text{Case 2: } n = 9$$

The smallest possible whole number is: 9.

Hence, option a is correct.

6. Which operators should be interchanged to get the minimum value of the given expression?

$$8 \times 4 + 2 - 1$$

- a)  $\times$  and +                      b) + and -                      c) - and  $\times$                       d) No change

**Answer:** c

**Solution:**

Let us solve the current expression:  $8 \times 4 + 2 - 1 = 32 + 2 - 1 = 33$ .

To find the minimum value by interchanging the operators, let's analyse the problem logically instead of trying each pair of operators given in the options.

Since we need the least possible result, we should consider changing the multiplication operator because it is placed between two of the highest numbers in the given expression.

By interchanging it with the subtraction operator, which is placed between two of the lowest numbers, we can reduce the overall value.

Let's apply this change to the expression: Original expression:  $8 \times 4 + 2 - 1$

Interchange multiplication with subtraction: New expression:  $8 - 4 + 2 \times 1$

Now, solve the new expression step by step:

$$2 \times 1 = 2$$

$$8 - 4 + 2 = 6$$

Thus, the minimum value is 6.

Option c is the correct choice.

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7. In certain code, 1 is coded as 2, 3 is coded as 10, 5 is coded as 26.

In the same way, 4 is coded as A and 6 is coded as B.

If C is the sum of A and B, which of the following is correct regarding C?

- a)  $C > 55$                       b)  $C < 50$                       c)  $C = 50$                       d)  $C < 55$

**Answer:** d

**Solution:**

Here, the pattern for coding is to multiply the number by itself and add 1. For example:

$$1 = (1 \times 1) + 1 = 2$$

$$3 = (3 \times 3) + 1 = 10$$

$$\text{So, } 4 = (4 \times 4) + 1 = 17$$

$$A = 17$$

$$\text{Also, } 6 = (6 \times 6) + 1 = 37$$

$$B = 37$$

Now,

$$C = A + B$$

$$C = 17 + 37 = 54$$

We know that  $54 < 55$

So,  $C < 55$

Therefore, the correct answer is option d,  $C < 55$ .

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8. In the following expression, each ball with a mathematical operator is swapped with the ball immediately to its left, to form a new arithmetic expression. What will be the value of NEW EXPRESSION - ORIGINAL EXPRESSION?

LEFT  $(8)(7)(+)(1)(5)(-)(3)$  RIGHT

Expression

- a) 63                      b) 73                      c) -63                      d) -73

**Answer:** d

**Solution:**

The value of the original expression is:

$$87 + 15 - 3$$

$$= 102 - 3$$

$$= 99$$

Let's form the new expression, as per the given instructions.

Given that, each ball with an operator is swapped with the ball immediately to its left.



After swapping the balls having mathematical operators with the balls on their immediate left, the resultant expression is:



$$= 8 + 71 - 53$$

$$= 79 - 53$$

$$= 26$$

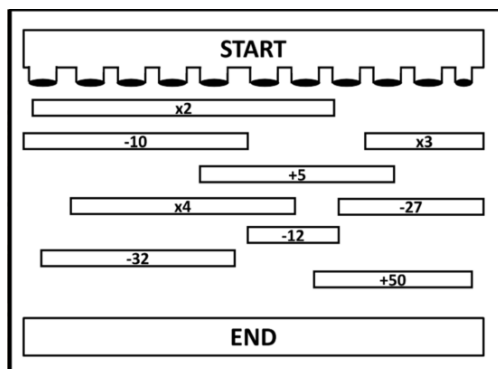
New expression - Original expression

$$= 26 - 99$$

$$= -73$$

Thus, the correct answer is option d.

9. A stone with an initial value of 10 is dropped through one of the holes at the **START** and moves in a straight path to reach the **END**. As it passes each numbered bar along the way, its value changes according to the operation shown on that bar. What is the maximum value the stone can have when it reaches the **END**?



a) 60

b) 88

c) 53

d) 68

**Answer:** b

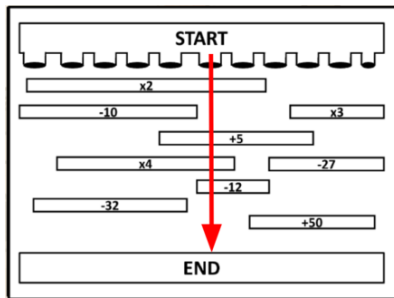
**Solution:**

As we need the highest possible value, we should try to touch any of these multiplication bars, so that there is a good increase in the stone's value.

However, if we concentrate on bar 'x3', we would miss the bars 'x2' and 'x4', which would be a great loss. So, let's ignore bar 'x3'.

We should avoid -32 and -10 as it would decrease the overall value.

Hence, we just take -12 over -32 and -10.



The value we get from the above hole is:

Bar 1:  $(10 \times 2) = 20$

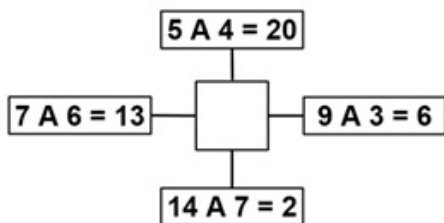
Bar 2:  $20 + 5 = 25$

Bar 3:  $25 \times 4 = 100$

Bar 4:  $100 - 12 = 88$

Hence, option b is the correct answer.

10. Each arrow has a mathematical operator. When placed in the centre box, its operator replaces the letter A in the equation it points to. Which arrow satisfies an equation based on this rule?



- a)  b)  c)  d) 

**Answer:** d

**Solution:**

Let us check each option one by one by replacing A with the operator shown on the arrow.

Option a: Operator = '-'

Top equation becomes:

$$5 - 4 = 1$$

Since  $1 \neq 20$ , this option is incorrect.

Option b: Operator = '÷'

Left equation becomes:

$$7 \div 6 = 1.16\dots$$

Since  $1.16 \neq 13$ , this option is incorrect.

Option c: Operator = '+'

Right equation becomes:

$$9 + 3 = 12$$

Since  $12 \neq 6$ , this option is incorrect.

Option d: Operator = '÷'

Bottom equation becomes:

$$14 \div 7 = 2$$

Since  $2 = 2$ , this equation holds true.

Therefore, the correct answer is option d.



## The Thinking Spot

Tom, Sam, Mariya, John, and Bob each selected a shape. They placed their shapes in the grid below.

Tom's shape is neither a circle nor a triangle

Sam's shape is in column number 3

Mariya's shape is not in the first three rows

John's shape is neither square nor circle

Which of the following shapes is chosen by Bob?

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | △ |   |   |   |   |
| 2 |   |   |   | ○ |   |
| 3 |   | □ |   |   |   |
| 4 |   |   | ☆ |   |   |
| 5 |   |   |   |   | ⬡ |

(a) Square

(b) Circle

(c) Hexagon

(d) Triangle

**Answer:** b

**Solution:**

- Sam: The shape in column 3 is a star.
- Mariya: Not in the first three rows, so it must be a hexagon because Sam has a star.
- Tom: Since it's not a circle or a triangle, it is a square because Sam has a star and Mariya has a hexagon.
- John: Since it's neither a square nor a circle, it has to be a triangle, the only shape left.
- Bob has selected a circle, which is the last remaining shape.

Hence, option b is the correct answer.

